Assignment #6. jQuery (Selectors, Effects, Animations)

Objective:

By the end of this assignment, students will be able to:

- find any element on a page with jQuery selectors and traversal methods;
- add, move and remove elements and show the changes with jQuery effects;
- make a page come alive with fades, flights and sliding cards.

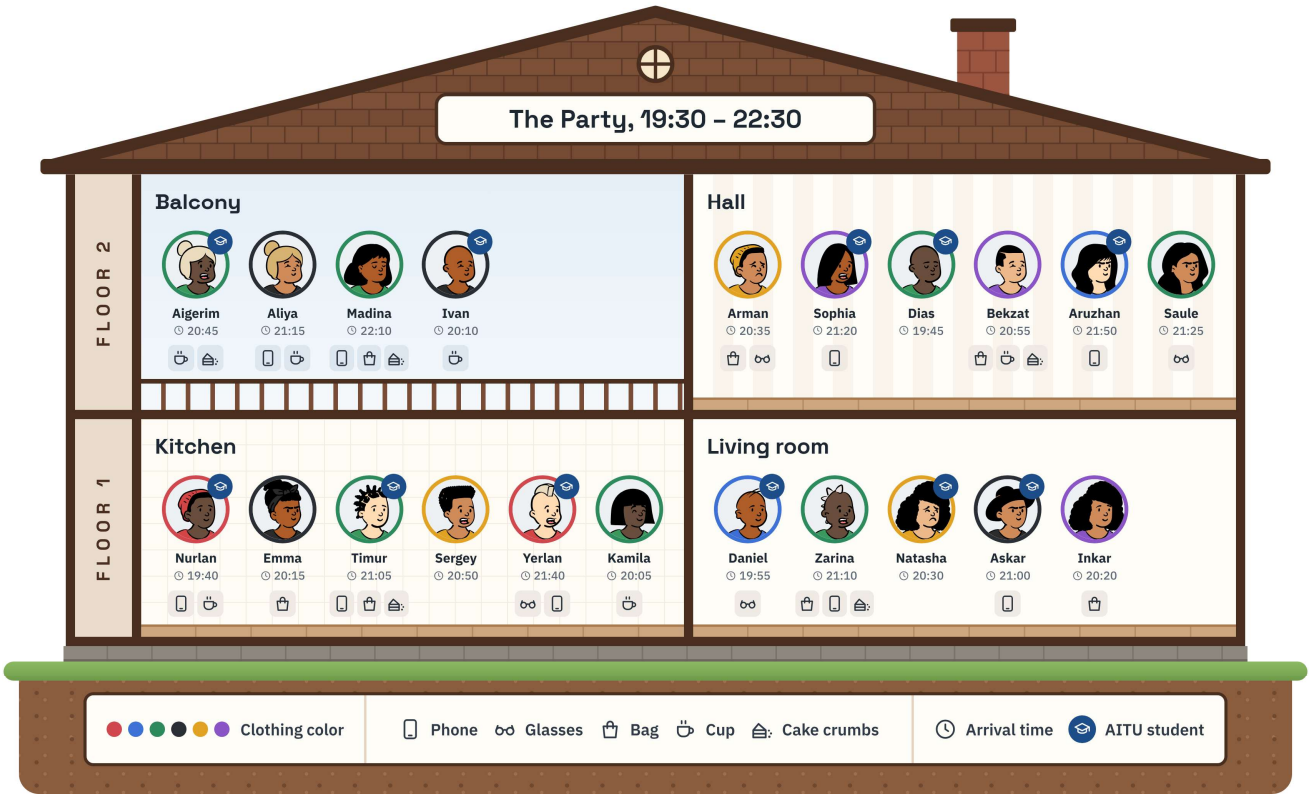
Rules for Submission:

1. This is an individual assignment. If the works of different students are the same, the grade is 0.
2. Submit before the deadline. **Late works are not accepted!** To get a grade you must **submit before the deadline AND defend your work at the practice lesson**. Submission without a defense will result in a grade of 0.
3. You must submit: your report in **Assignment6_YourName.docx** (or .pdf) format. If you submit in other formats, your grade will be **0**. All source code must be uploaded to the GitHub repository.
4. If you cannot explain your own code, the whole assignment is graded 0.
5. Use jQuery for all the behavior: no other libraries, no onclick attributes, no CSS transitions or CSS animations.
6. IN THE REPORT MUST BE:
   - Your name and group
   - Every task with screenshots, tables and a short explanation in your own words.
   - A short conclusion about what you learned
   - The GitHub repository URL with the source code. Submissions without the link will receive a grade of 0.

Task 1. Selector Detective (selectors and traversal)

AITU students met at a birthday party in a house. The guests were in 4 rooms from 19:30 to 22:30. At 21:30 the birthday cake disappeared from the kitchen. The picture shows everyone who was there: where they stood, what they wore, when they came and what they had with them. Rebuild the party on a web page and find the thief with jQuery.

1. Write the party from the picture as an array: rooms with their name and floor, and guests from left to right with the name, clothing color, arrival time, AITU student or not, and things.
2. Build the page from the array with jQuery: 2 floors with rooms, and a card for every guest with the avatar, a ring in the clothing color, name, time, icons of the things and an AITU badge. Store the guest data on the card as classes and data-* attributes. The design is yours.
3. Add a button for every case. A click highlights what the case finds, makes the found guests jump once with **animate()** and shows their number and names.
4. Solve every case with one jQuery chain that searches the page, not the array. Before every case, build the page again; cases 18 and 19 change the party with **fadeIn()** and **fadeOut()**.
5. In your report, show every case: its jQuery chain and a screenshot of the result on the page.



Avatars: use any free avatars, for example Open Peeps (free, CC0).

An avatar for a name is one link: https://api.dicebear.com/9.x/open-peeps/svg?seed=Nurlan

Cases

Case	Find or do
1	Who is in the hall?
2	Who is dressed in yellow or purple?
3	Which AITU students are upstairs?
4	Who has glasses or a bag?
5	Who came with nothing at all?
6	Who has a phone but no bag?
7	Who arrived between 20:30 and 21:00, both times included?
8	Who arrived after 21:00 and is not an AITU student?
9	Who stands right next to Askar?
10	Who stands last in each room?
11	Who stands third in each room?
12	Who stands after Timur in his room?
13	Who stands right before someone in black?
14	In which room is Aruzhan?
15	Which rooms have fewer than 6 guests?
16	In which rooms does somebody have a cup, but nobody wears glasses?
17	Who came first to each room?
18	Miras, an AITU student in blue, comes at 21:20 with his phone and joins Askar and Inkar, standing between them. Let him appear smoothly.
19	Natasha says goodbye and goes home. Her card fades away and disappears from the page.
20	**Who took the cake? The thief was downstairs, came after 21:00 but before the cake disappeared, has crumbs but no cup, and stood right next to an AITU student.**

Tools: id, class and attribute selectors, **:has()**, **:not()**, **:last-child**, **:nth-child()**, **filter()**, **next()**, **prev()**, **nextAll()**, **siblings()**, **closest()**, **add()**, **insertAfter()**, **fadeIn()**, **fadeOut()**, **remove()**.

Task 2. The party in motion (effects and animations)

Create a copy of your Task 1 files and continue in the copy. Make the page come alive with effects.

1. **Spotlight.** Click a case. The guests that the case finds stay bright. All other guests become pale: they fade to 20% in 0.3 seconds. Some cases find rooms, not guests. For these cases, the guests in all the other rooms become pale. The Clear button makes all guests bright again.
2. **Guest card.** Click a guest. A card with all the information about this guest opens. The card shows the information in words, for example “Came at 20:30”, “AITU student”, “Has a phone and a bag”. The card does not just appear: it starts small on the guest’s photo and grows to its full size. The card closes when the user presses Esc or clicks outside the card. Then it gets small again and goes back into the photo. Only one card can be open. When the user clicks another guest, the first card closes and the new one opens.
3. **Move.** The guest card has a button for each other room, for example “Move to the hall”. Click a button to send the guest to the end of that room. The guest’s card flies across the house to the new room; it does not jump. In the old room, the empty place closes slowly. In the new room, a place opens slowly. After the move, change your array too, so that the cases from Task 1 find the guest in the new room.
4. **Sort by time.** Each room has a button “Sort by time”. Click it, and the guests of this room stand in order of their arrival time, from the earliest to the latest. Click it again, and the guests go back to the order in the picture. The cards move slowly to their new places; they do not jump.
5. **Fast clicks.** Click the buttons and guests very fast, many times. Every effect must still end in the right state. No animation repeats again and again after the user stops clicking.

Check	**Expected result**
Case 3, then Clear	Only the AITU students on the 2nd floor stay bright; Clear brings everybody back.
Click a guest, then Esc	The card grows out of the avatar with all the data; Esc shrinks it back.
Move Arman to the balcony	Arman flies to the end of the balcony; the hall closes the gap; case 1 no longer finds him.
Sort the living room twice	The cards slide to 19:55, 20:20, 20:30, 21:00, 21:10, then back to the picture order.
Sort 5 times fast	The cards end in the sorted order without jumping or shaking.

Tools: fadeTo(), **animate()** with callbacks, **stop()**, **offset()**, **position()**, **clone()**, **css()**, **append()**, **on()** with click and keydown.

Assessment criteria

Criterion	What is checked	Points
Task 1	- the array matches the picture; the page is built from it - every case gives the right result with one jQuery chain	22
Task 2	- every check in the table gives the shown result - the four effects are animated with jQuery; fast clicks leave no queues	22
Report	- contains everything listed in the rules; explanations in the student's own words	6
Defense	- explains the work and demonstrates it live without errors (10) - answers the instructor's questions about their own code and the topic (15) - completes the live coding task independently, without assistance or hints (25)	50
Total		100

Learn more:

https://www.w3schools.com/jquery/jquery_ref_selectors.asp

https://www.w3schools.com/jquery/jquery_traversing.asp

https://www.dicebear.com/styles/open-peeps/

https://www.w3schools.com/jquery/jquery_animate.asp

https://www.w3schools.com/jquery/jquery_fade.asp

https://www.w3schools.com/jquery/jquery_dimensions.asp

Good luck!